

EDS 223 HW 1: Map making practice

AUTHOR

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Load packages

```
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(janitor)
library(kableExtra)
library(dplyr)
```

Load data

```
# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data", "ejscreen", "EJSCREEN_2023_BG_StatePct_with_AS_
  clean_names()
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
`/Users/emaliapartlow/Desktop/MEDS/eds223/eds223-
hw1/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:      XY
Bounding box:   xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS:  WGS 84 / Pseudo-Mercator
```

Initial data exploration

```
# Check dataframe dimensions
dim(ejscreen)
```

```
[1] 243021    224
```

```
# Look at first 3 rows
head(ejscreen, n = 3) |>
  kable()|>
  kable_styling()
```

id	state_name	st_abbrev	cnty_name	region	acstotpop	acsipovbas	acseducbas	acstothh	acs
010010201001	Alabama	AL	Autauga County	4	693	693	489	252	
010010201002	Alabama	AL	Autauga County	4	1098	1098	736	357	
010010202001	Alabama	AL	Autauga County	4	844	808	523	272	

Filter to Monterey county

```
# Filter to Monterey
monterey <- ejsscreen |>
  dplyr::filter(st_abbrev == 'CA') |>
  dplyr::filter((cnty_name == 'Monterey County'))

# Check new dimensions
dim(monterey)
```

```
[1] 272 224
```

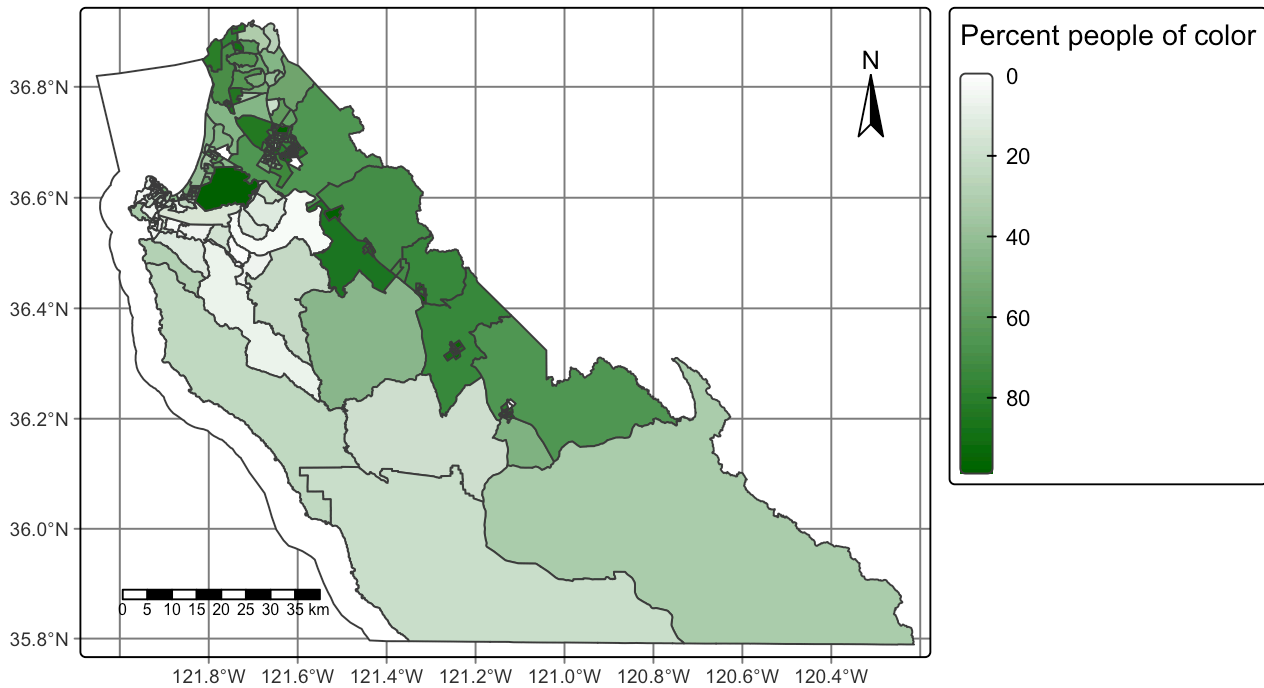
```
# Check that filtered to Monterey correctly
monterey |>
  distinct(cnty_name)
```

```
      cnty_name
1 Monterey County
```

Map of people of color in Monterey

```
tm_shape(monterey) +
tm_graticules() +
  tm_polygons(
    fill = 'p_peopcolorpct',
    fill.scale = tm_scale_continuous(values = c('white', 'darkgreen')),
    fill.legend = tm_legend(title = 'Percent people of color',
                             position = tm_pos_out('right', 'center'))
  ) +
tm_compass(position = c('right', 'top')) +
tm_scalebar(position = c('left', 'bottom')) +
tm_title('People of color in Monterey') +
tm_legend(position = tm_pos_out('right', 'center'))
```

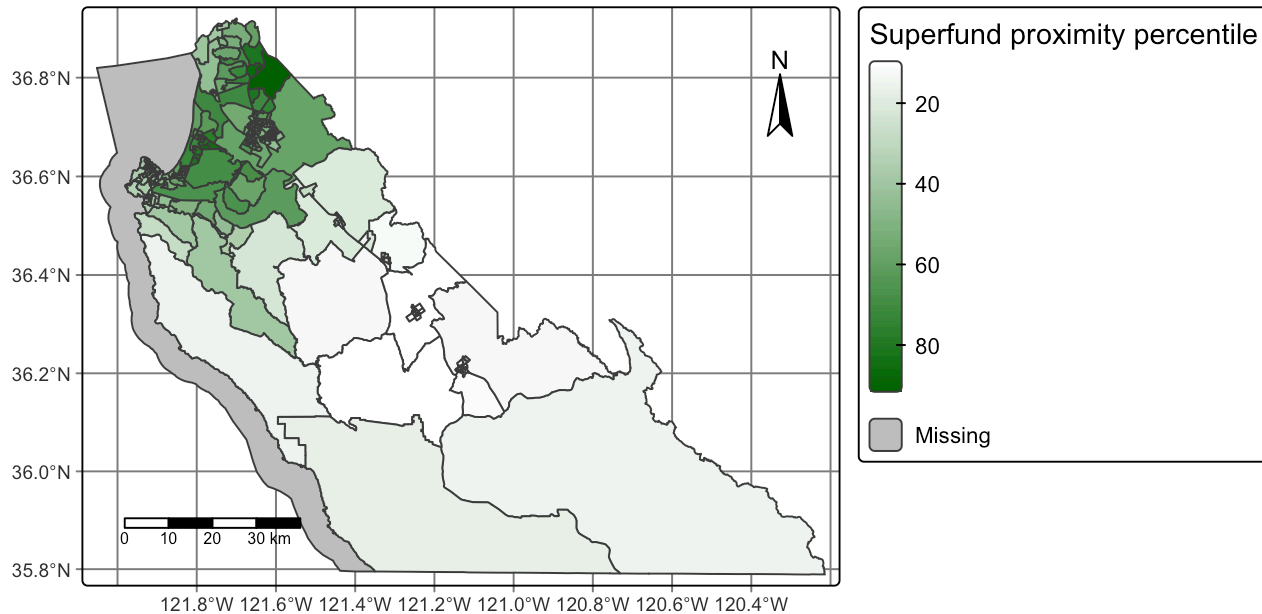
People of color in Monterey



Map of Superfund proximity in Monterey

```
tm_shape(monterey) +
tm_graticules() +
  tm_polygons(
    fill = 'p_pnpl',
    fill.scale = tm_scale_continuous(values = c('white', 'darkgreen')),
    fill.legend = tm_legend(title = 'Superfund proximity percentile',
                           position = tm_pos_out('right', 'center'))
  ) +
tm_compass(position = c('right', 'top')) +
tm_scalebar(position = c('left', 'bottom')) +
tm_title('Superfund proximity in Monterey') +
tm_legend(position = tm_pos_out('right', 'center'))
```

Superfund proximity in Monterey



Map interpretation

The percent proximity to a Superfund site is highest at the northern end of Monterey county, which is also where there is the highest percentage of people of color. The areas closest to a Superfund site are consistent with the location of Fort Ord, a former U.S Army base that was established as a Superfund site by the EPA in 1990. These maps display that people of color may be more exposed to Fort Ord's contaminated soil, groundwater, and unexploded ordinances due to their close proximity to the Superfund site.